

Kunwoo Park

Ph.D. Student, Department of Electrical and Computer Engineering
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Research Interests

Optical computing; disordered photonics; topological photonics; open quantum systems.

Education

Integrated M.S./Ph.D., Department of Electrical and Computer Engineering 2024.09 – present
Seoul National University Advisor: Sunkyu Yu (Expected: 2029.08)

B.S., Department of Electrical and Computer Engineering 2018.03 – 2024.08
Seoul National University

Military Service, Republic of Korea Army 2020.12 – 2022.06

Academic Positions

- Placeholder.

Research Experience

- Optical Computing.** Placeholder.
- Disordered Photonics.** Placeholder.
- Topological Photonics.** Placeholder.
- Open Quantum Systems.** Placeholder.

Publications

Preprints

- Gitae Lee, Seungmok Youn, Ikbeom Lee, **Kunwoo Park**, Duwhan Hwang, Xianji Piao^{*}, Namkyoo Park^{*} and Sunkyu Yu^{*}. Non-Hermitian stealthy hyperuniformity. [arXiv:2602.09458](#).
- Kunwoo Park**, Ikbeom Lee, Seungmok Youn, Gitae Lee, Namkyoo Park^{*} and Sunkyu Yu^{*}. Hypergraph modelling of wave scattering to speed-up material design. [arXiv:2507.15329](#).

Journal Articles

- Seungmok Youn, **Kunwoo Park**, Ikbeom Lee, Gitae Lee, Namkyoo Park^{*} and Sunkyu Yu^{*}. Phase-sensitive engineering of optical disordered materials using heterogeneous networks. *Advanced Optical Materials* 14, e03499 (2026). [doi:10.1002/adom.202503499](#).
- Kyuhoo Kim[†], **Kunwoo Park**[†], Hyungchul Park, Sunkyu Yu, Namkyoo Park^{*} and Xianji Piao^{*}. Programmable photonic unitary circuits for light computing. *Nanophotonics* 14, 1429 (2025). [doi:10.1515/nanoph-2024-0602](#).

Conference Proceedings

- Ikbeom Lee, **Kunwoo Park**, Seungmok Youn and Sunkyu Yu^{*}. Inverse design of microstructures of non-Hermitian scattering systems. SPIE Optics + Photonics 2025, 13578-59, San Diego, California, USA, 6 August 2025. [doi:10.1117/12.3062322](#).
- Kunwoo Park**, Ikbeom Lee, Seungmok Youn and Sunkyu Yu^{*}. Modularization of light scattering for photonic computing applications. SPIE Optics + Photonics 2025, 13581-25, San Diego, California, USA, 5 August 2025. [doi:10.1117/12.3062195](#).
- Ikbeom Lee, **Kunwoo Park** and Sunkyu Yu^{*}. Design of wave scattering from gain and loss disordered materials. CLEO-PacificRim (CLEO-PR) 2024, Mo2J-4, Incheon, Republic of Korea, 5 August 2024. [doi:10.1109/CLEO-PR60912.2024.10676414](#).
- Kunwoo Park**, Ikbeom Lee and Sunkyu Yu^{*}. Inverse design method of hyperuniform materials using system factorization. CLEO-PacificRim (CLEO-PR) 2024, Mo2J-2, Incheon, Republic of Korea, 5 August 2024.

doi:10.1109/CLEO-PR60912.2024.10676648.

Patents

1. Nonlinear unitary optical device. Sunkyu Yu, Namkyoo Park, Xianji Piao and **Kunwoo Park**. KR 10-2859166; US application 19/048,370; EU application 25156862.2.

Grants and Fellowships

Graduate Research Encouragement Grant () 2025 – present
Principal Investigator

Teaching

Application of Quantum Mechanics (Graduate), Teaching Assistant Spring 2025
Seoul National University

Topics in Electro-physics (Graduate), Teaching Assistant Spring 2025
Seoul National University

Technical Skills

- **Theory and modeling:** Placeholder.
- **Computation:** Placeholder.
- **Experimentation:** Placeholder.
- **Software:** Placeholder.

Professional Service

- Placeholder.

References

Available upon request.

{[†] Co-first authors. * Corresponding authors.}